

Graphing Free Fall: Position, Velocity and Acceleration

SimuPhysics Free Fall Simulation and Graphs · SimuPhysics

Year 10

~30 minutes

Science · Physics

NOTICE — AI-generated worksheet

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Simulation: <https://simuphysics.com/view/mechanics-free-fall-simulation-and-graphs.html>

Curriculum links: AC9S10U05 · AC9S7U04

Learning intentions

- I can sketch and interpret position-time, velocity-time and acceleration-time graphs for a falling object.
- I can explain how gravity produces constant acceleration during free fall.
- I can quantitatively relate force, mass and acceleration for a falling object using Newton's laws.

Getting started

Open the SimuPhysics Free Fall Simulation and Graphs tool. Before you drop the object, sketch on paper what you think the position-time, velocity-time and acceleration-time graphs will look like. Then drop the object and compare your sketches with the simulation's live graphs.

Tasks

1. Drop the object and describe what happens to its speed as it falls, based on the velocity-time graph.

2. Compare your sketched graphs from before the drop to the simulation's actual graphs. What did you get right, and what surprised you?

3. Look at the acceleration-time graph while the object is falling. What shape is it? What does this tell you about the object's acceleration?

4. If the object is in free fall for 3 seconds and gravity gives it an acceleration of about 9.8 m/s^2 , calculate its final velocity, assuming it started at rest.

Working:

5. Using your answer from the previous task, estimate how far the object fell in those 3 seconds (average velocity $\times$ time).

Working:

6. A heavier object and a lighter object are dropped from the same height with no air resistance. Use Newton's second law (force = mass $\times$ acceleration) to explain why they hit the ground at the same time.

7. Compare the position-time graph's shape (a curve) to the velocity-time graph's shape (a straight line). Explain why they look different even though they describe the same motion.

8. In real life, a falling leaf does not accelerate at a constant 9.8 m/s^2 . Explain what force this simulation may be leaving out, and how it would change the graphs.

Extension (optional)

Estimate how long it would take an object to fall from a 100 m building, ignoring air resistance. Show your working, then explain one reason why a real object dropped from that height might take slightly longer.
